



Loop a Pattern in Code

Ontario Math C3.1 — repeating events

Name: _____

Date: _____

★ I can loop a small pattern of moves to make a staircase.

Repeat a little pattern

A staircase is the same step over and over. Bit's step is: go across, then go up. A loop repeats that step to build the whole staircase.

Read the code — `range(3)` repeats the across-then-up step three times.

Read this Python code

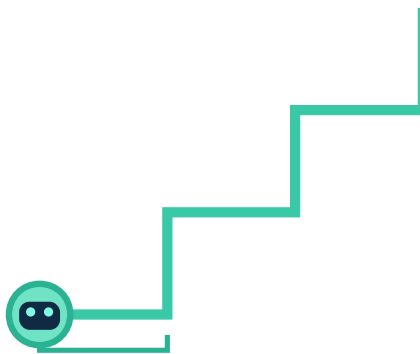
PYTHON

```
for step in range(3):  
    bit.forward(30)  
    bit.left(90)  
    bit.forward(30)  
    bit.right(90)
```

OUTPUT

Bit climbs a 3-step staircase.

`range(3)` → 3 steps up



Each repeat is one step: across, then up. `range(3)` makes 3 steps.

1 Read it

How many steps does the staircase have? _____ Each step has 2 forward moves, so how many forward moves in TOTAL? _____

2 Predict, then run

You change `range(3)` to `range(5)`. PREDICT: how many steps, and how many forward moves in total? Then run it.

3 Write your own

Make Bit climb a 4-step staircase. Write the range number: for step in `range(_____)`:

4 Challenge

Each step goes UP 30. If Bit climbs a 4-step staircase, how high does Bit go in total? Show your thinking.

Big idea

A loop can repeat a little PATTERN of moves, not just one move. Repeat the across-then-up step and Bit builds a whole staircase.



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Quick facilitation notes & answer key

What this worksheet teaches

A staircase is the same little step over and over: go across, then go up. Students read a repeat that does that whole step several times to build a staircase, and count the moves and total height. A repeat can repeat a whole pattern, not just one move. No coding experience needed.

Before you start — showing the drawing (optional)

The worksheet shows the typed instructions AND a picture of the staircase Bit climbs, so you can teach straight from the printout — no computer required.

To show it live (optional): open a browser, go to a free "Python turtle" playground (e.g. search "trinket python turtle"), type the few lines from the worksheet, and press Run — a turtle draws it. You only press Run.

How to lead it (about 30 minutes)

1. Walk a staircase as a class — across, up, across, up — and name the repeating step.
2. Show it (you do). Run the 3-step staircase; trace one step, then the repeat. Graph paper helps students sketch it.
3. Let them work. Students do Exercises 1–4 (Ex 4 multiplies the rise), solo or in pairs.
4. Wrap up: a repeat can repeat a whole pattern of moves, not just one.

Answer key (these are the verified correct answers)

Exercise 1 — 3 steps; 6 forward moves (2 per step $\times$ 3).

Exercise 2 — 5 steps; 10 forward moves in total.

Exercise 3 — repeat 4 times.

Exercise 4 (challenge) — 120 high (4 steps $\times$ 30 up).

If students get stuck — and ways to adjust

Watch for: counting one forward per step instead of two.

Make it easier: highlight the two "move forward" lines inside the repeat before counting.

Make it harder: ask the height of a 6-step staircase (180).

Ontario curriculum link (official wording)

C3.1 — solve problems and create computational representations of mathematical situations by writing and executing code, including code that involves sequential, concurrent, and repeating events.

In plain terms: students write and run step-by-step instructions — including a "repeat" instruction — to make something happen.

C3.2 — read and alter existing code, including code that involves sequential, concurrent, and repeating events, and describe how changes to the code affect the outcomes.

In plain terms: students read instructions, change one, and explain what the change did to the result.

You'll know it worked when

The student counts steps and total forwards (Ex 1–2), writes the repeat number for a 4-step staircase (Ex 3), and multiplies to find the total height (Ex 4).